Net force first

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Learning target: Combine forces before using mass to predict acceleration.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 10.*
Earth pulls the sign down with 50 N; the cord pulls it up with 50 N.

02. Worked example: explain the model
Net arrow 6 N right; 9 N omits the opposing force.

03. Read the arrows
12-4=8 N right; 8/4=2 m/s² right.

04. Balanced but moving
Net 0 N; a 0; keeps same leftward velocity.

05. Compare two masses
3 m/s² and 2 m/s²; 2 kg cart.

06. Review: connect an earlier idea *Review: Unit 1, Day 1.*
68 m if the 7 m/s rightward motion continues. Check a later timed position against the prediction.

07. Exit check
32 N right; 4 m/s² right.

08. STRETCH: test the model
Examples 8 and 2 N; 11 and 5 N. Both give 3 m/s² right.

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01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 10.*
Weight 39.2 N down from Earth and tension 39.2 N up from string. Object-on-string acts on the string, not the object.

02. Worked example: explain the model
Net arrow 6 N right; 9 N omits the opposing force.

03. Work from force arrows
Net 20 N right; a 4 m/s² right; v=-6+4=-2 m/s(left), slowing.

04. Find an unknown force
4-L=3(-2); L=10 N.

05. From braking to force
Forward positive; a=-6 m/s²; F=-9000 N.

06. Review: connect an earlier idea *Review: Unit 1, Day 1.*
v=-3.5 m/s; x(9)=-10.5 m. Assumes constant velocity; compare another timed position with its uncertainty.

07. Exit check
Net -12 N; a=-3 m/s². Need current velocity; leftward motion speeds up, rightward motion slows.

08. STRETCH: test the model
Net -6 N means other force -16 N. It can be moving right and slowing.